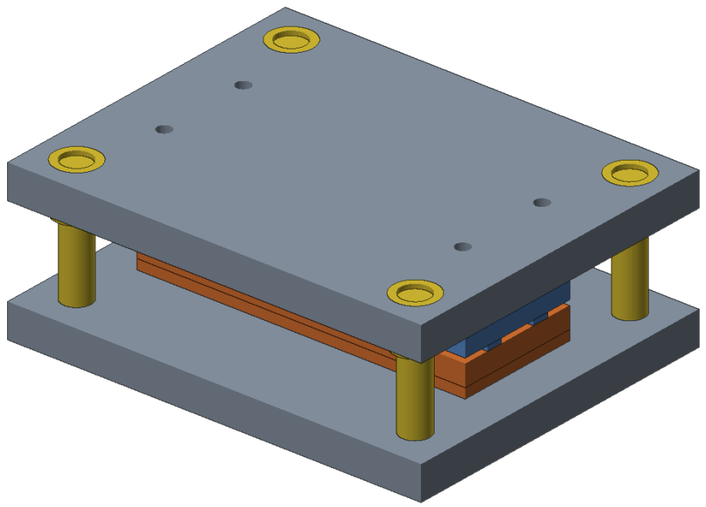
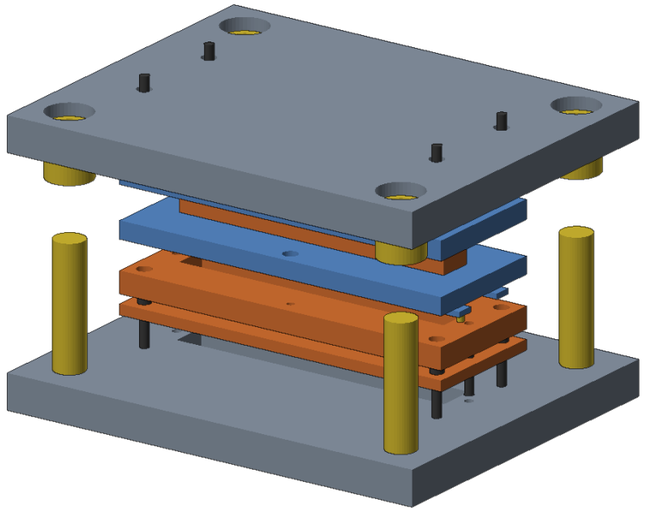
Design to Cost · TOOL ASSEMBLY

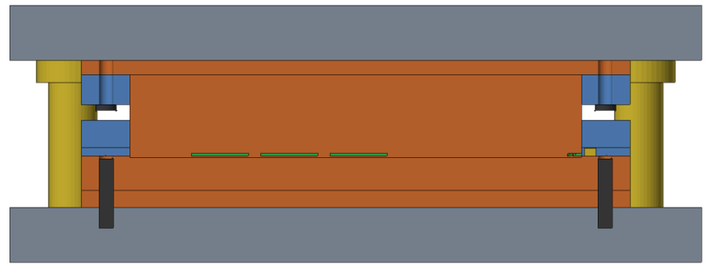
EX26-0001 · Die 1: Blanking



ISO VIEW (closed, bottom of stroke)



EXPLODED VIEW



SECTION A-A (through the middle, from the front)

Item	Qty	Description	Material / grade	Hardness	Make / standard part	Size (mm)
01	1	Bottom shoe (die set)	C45 / S50C	-	Local die-set maker / Misumi	500.0 × 400.0 × 40.0
02	1	Top shoe (die set)	C45 / S50C	-	Local die-set maker / Misumi	500.0 × 400.0 × 40.0
03	1	Die plate	D2 (1.2379)	58-62 HRC	-	396.7 × 150.0 × 25.0
04	1	Backing plate (die)	Mould base steel (4140 / 1.722	28-32 HRC pre-hardened	-	396.7 × 150.0 × 12.0
05	2	Stock guide	C45 / S50C	annealed	-	396.7 × 20.0 × 6.0
06	1	Stripper plate (fixed)	C45 / S50C	annealed	-	396.7 × 150.0 × 20.0
07	1	Punch holder plate	C45 / S50C	annealed	-	396.7 × 150.0 × 22.0
08	1	Backing plate (punch)	Mould base steel (4140 / 1.722	28-32 HRC pre-hardened	-	396.7 × 150.0 × 10.0
09	1	Blanking punch	D2 (1.2379)	58-62 HRC	-	325.9 × 39.8 × 60.0
10	1	Stop pin Ø8	-	-	Stop pin ISO 8734 Ø8 m6 × 18	8.0 × 8.0 × 18.0
11	4	Guide pillar Ø35	-	-	Guide pillar Ø35 g6 × 185 (die set 4-pillar die set	35.0 × 35.0 × 181.0
12	4	Guide bush Ø35	-	-	Guide bush Ø35 H7 × OD 52.5 × 56	52.5 × 52.5 × 56.0
13	4	Socket head cap screw M10 × 45	-	-	ISO 4762 M10 × 45 - 12.9	15.0 × 15.0 × 55.0
14	2	Dowel pin Ø10 × 50	-	-	ISO 8734 Ø10 m6 × 50	10.0 × 10.0 × 50.0
15	4	Socket head cap screw M10 × 55 (top)	-	-	ISO 4762 M10 × 55 - 12.9	15.0 × 15.0 × 65.0

Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Closed (bottom of stroke): shut height 186.0 mm; strip 46.0 mm wide between the stock guides. · 3. Press PRESS-100 for 61.7 t. · 4. Fits: pillar g6, bush H7, dowel m6 / H7, punch m6 / H7

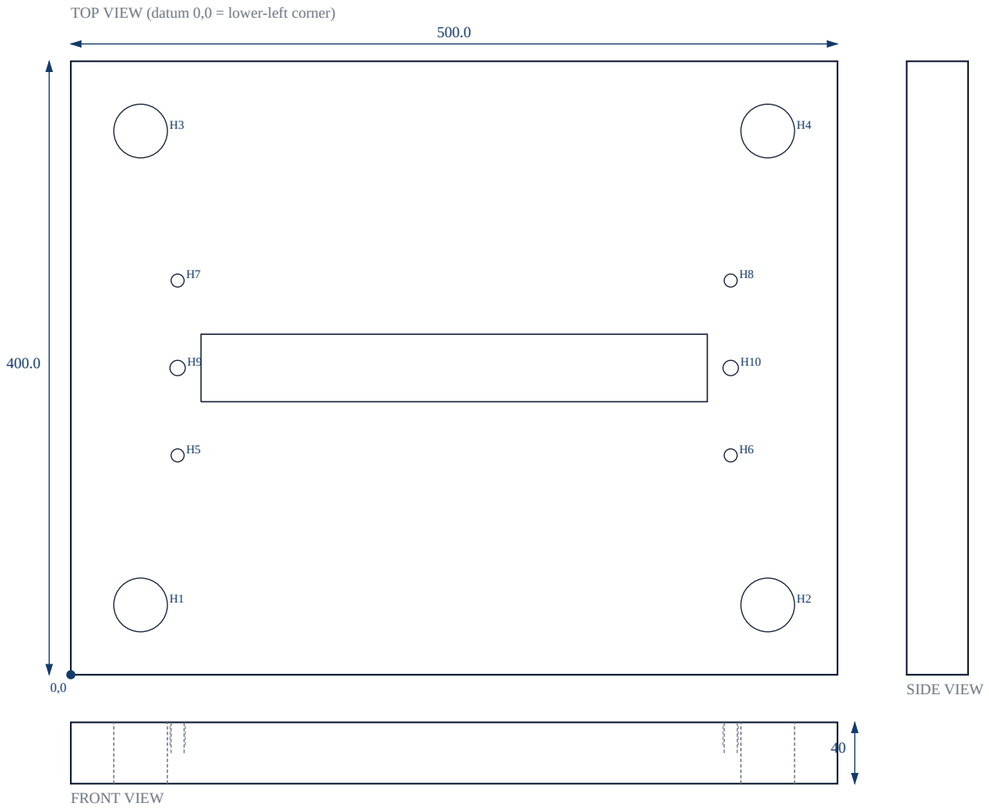
Design to Cost
Drawing no. **EX26-0001-T01-00**
Rev **B**

Part	EX26-0001 · L mounting bracket	**Die set / base**	4-pillar die set 5040 500 × 400
Tool	Die 1: Blanking	**Shut height**	186 mm

General tolerance	ISO 2768-mK
Projection	
Scale	as shown
Sheet	1 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

01 · Bottom shoe (die set)



#	X	Y	Ø	Kind / fit
H1	45.50	45.50	35	Ø35 H7 pillar seat, position ±0.01
H2	454.50	45.50	35	Ø35 H7 pillar seat, position ±0.01
H3	45.50	354.50	35	Ø35 H7 pillar seat, position ±0.01
H4	454.50	354.50	35	Ø35 H7 pillar seat, position ±0.01
H5	69.65	143.00	8.5	M10 tapped, depth 20
H6	430.35	143.00	8.5	M10 tapped, depth 20
H7	69.65	257.00	8.5	M10 tapped, depth 20
H8	430.35	257.00	8.5	M10 tapped, depth 20
H9	69.65	200.00	10	Ø10 m6 / H7 dowel, depth 15, position ±0.01
H10	430.35	200.00	10	Ø10 m6 / H7 dowel, depth 15, position ±0.01

Slug escape · profile in the DXF (wire cut / CNC)

Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Part of die set 4-pillar die set 5040

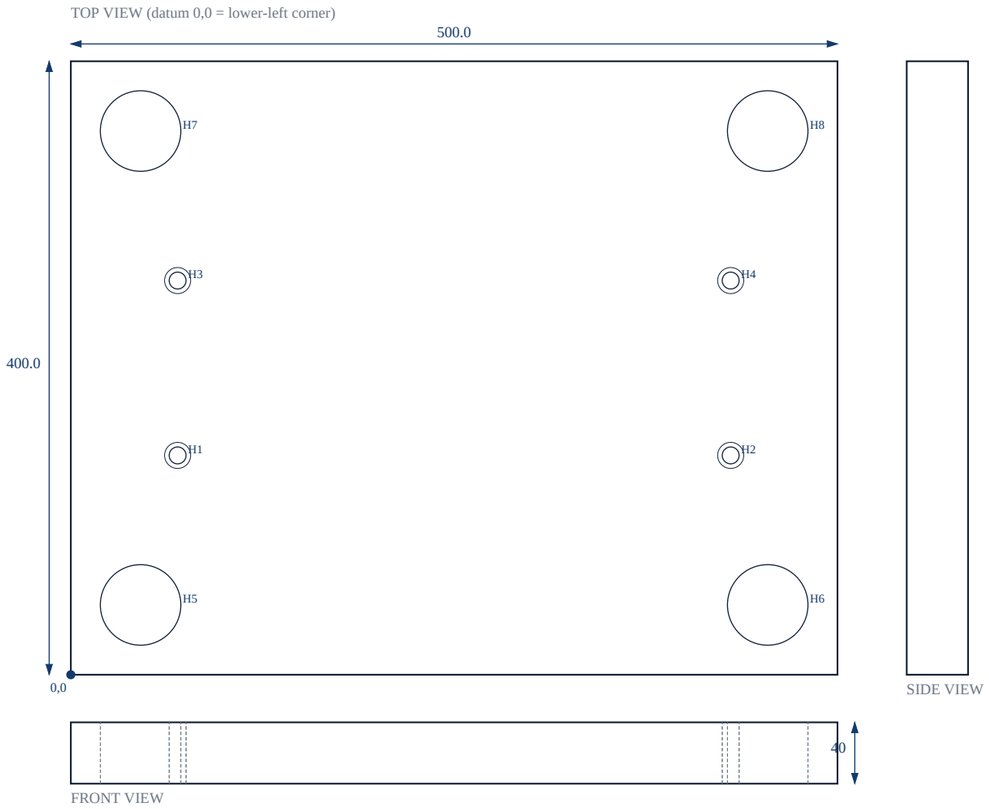
Design to Cost
Drawing no. **EX26-0001-T01-01**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	-
Description	Bottom shoe (die set)	**Size**	500.00 × 400.00 × 40.00 mm
Item / qty	01 · 1 off	**Make**	Local die-set maker / Misumi
Material	C45 / S50C	**Finish**	Ra 1.6

General tolerance	ISO 2768-mK
Projection	
Scale	1:4
Sheet	2 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

02 · Top shoe (die set)



#	X	Y	Ø	Kind / fit
H1	69.65	143.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H2	430.35	143.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H3	69.65	257.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H4	430.35	257.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H5	45.50	45.50	52.5	Ø52.5 H7 bush seat, position ±0.01
H6	454.50	45.50	52.5	Ø52.5 H7 bush seat, position ±0.01
H7	45.50	354.50	52.5	Ø52.5 H7 bush seat, position ±0.01
H8	454.50	354.50	52.5	Ø52.5 H7 bush seat, position ±0.01

Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Part of die set 4-pillar die set 5040

Design to Cost
Drawing no. **EX26-0001-T01-02**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	-
Description	Top shoe (die set)	**Size**	500.00 × 400.00 × 40.00 mm
Item / qty	02 · 1 off	**Make**	Local die-set maker / Misumi
Material	C45 / S50C	**Finish**	Ra 1.6

General tolerance	ISO 2768-mK
Projection	
Scale	1:4
Sheet	3 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

03 · Die plate

#	X	Y	Ø	Kind / fit
H1	18.00	18.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H2	378.70	18.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H3	18.00	132.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H4	378.70	132.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H5	18.00	75.00	10	Ø10 m6 / H7 dowel, position ±0.01
H6	378.70	75.00	10	Ø10 m6 / H7 dowel, position ±0.01
H7	198.35	18.00	8.5	M10 tapped (stripper), depth 20
H8	198.35	132.00	8.5	M10 tapped (stripper), depth 20
H9	367.84	75.00	8	Ø8 m6 / H7 stop pin, depth 12, position ±0.01

Die opening = blank outline (wire cut) · tolerance +0.01 / 0 · profile in the DXF (wire cut / CNC)

TOP VIEW (datum 0,0 = lower-left corner)



SIDE VIEW

FRONT VIEW

Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Harden and temper to the hardness of the grade; stress relieve before wire cut / EDM finish · 6. Working faces Ra 0.4 · 7. Die opening +0.01 / 0 mm, land 4 mm then 1° relief · 8. Cutting clearance 0.120 mm per side (6 % of t)

Design to Cost
Drawing no. **EX26-0001-T01-03**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	58-62 HRC
Description	Die plate	**Size**	396.70 × 150.00 × 25.00 mm
Item / qty	03 · 1 off	**Make**	-
Material	D2 (1.2379)	**Finish**	Ra 1.6

General tolerance	ISO 2768-mK
Projection	
Scale	1:2
Sheet	4 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

04 · Backing plate (die)

#	X	Y	Ø	Kind / fit
H1	18.00	18.00	11	Ø11 screw clearance
H2	378.70	18.00	11	Ø11 screw clearance
H3	18.00	132.00	11	Ø11 screw clearance
H4	378.70	132.00	11	Ø11 screw clearance
H5	18.00	75.00	10	Ø10 m6 / H7 dowel, position ±0.01
H6	378.70	75.00	10	Ø10 m6 / H7 dowel, position ±0.01

Slug relief · profile in the DXF (wire cut / CNC)



Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Harden and temper to the hardness of the grade; stress relieve before wire cut / EDM finish

Design to Cost
Drawing no. **EX26-0001-T01-04**
Rev **B**

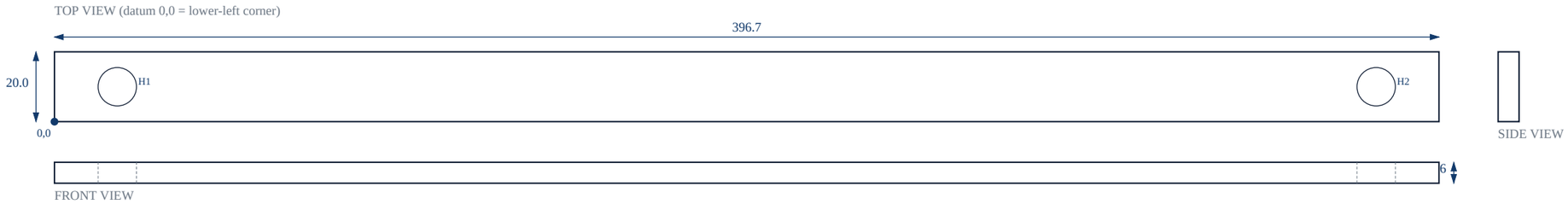
Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	28-32 HRC pre-hardened
Description	Backing plate (die)	**Size**	396.70 × 150.00 × 12.00 mm
Item / qty	04 · 1 off	**Make**	-
Material	Mould base steel (4140 / 1.7225)	**Finish**	Ra 1.6

General tolerance	ISO 2768-mK
Projection	
Scale	1:2
Sheet	5 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

05 · Stock guide

#	X	Y	Ø	Kind / fit
H1	18.00	10.00	11	Ø11 screw clearance
H2	378.70	10.00	11	Ø11 screw clearance



Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Strip width 46.00 mm + 0.1 mm gap per side

Design to Cost
Drawing no. **EX26-0001-T01-05**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	annealed
Description	Stock guide	**Size**	396.70 × 20.00 × 6.00 mm
Item / qty	05 · 2 off	**Make**	-
Material	C45 / S50C	**Finish**	Ra 1.6

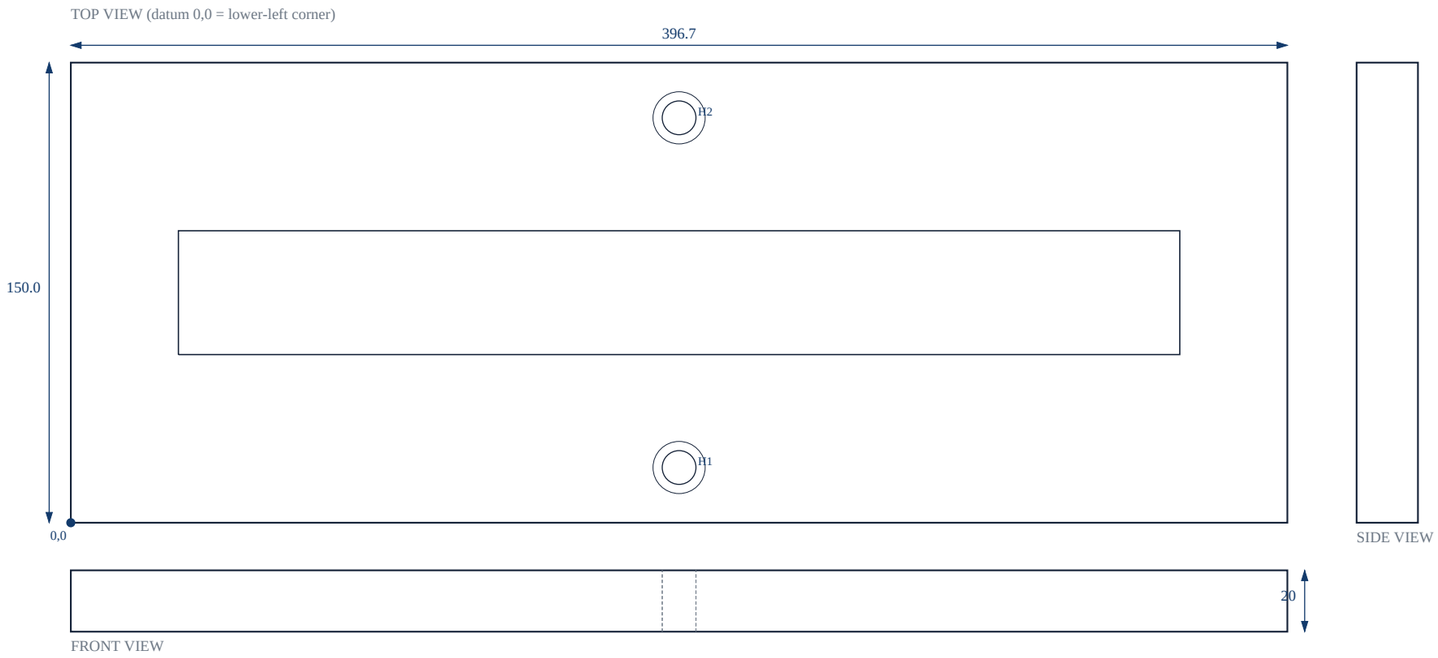
General tolerance	ISO 2768-mK
Projection	
Scale	1:1.5
Sheet	6 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

06 · Stripper plate (fixed)

#	X	Y	Ø	Kind / fit
H1	198.35	18.00	11	M10: Ø11 thru, counterbore Ø17 × 11
H2	198.35	132.00	11	M10: Ø11 thru, counterbore Ø17 × 11

Punch opening + gap · profile in the DXF (wire cut / CNC)



Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Punch openings + 0.3 mm per side

Design to Cost
Drawing no. **EX26-0001-T01-06**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	annealed
Description	Stripper plate (fixed)	**Size**	396.70 × 150.00 × 20.00 mm
Item / qty	06 · 1 off	**Make**	-
Material	C45 / S50C	**Finish**	Ra 1.6

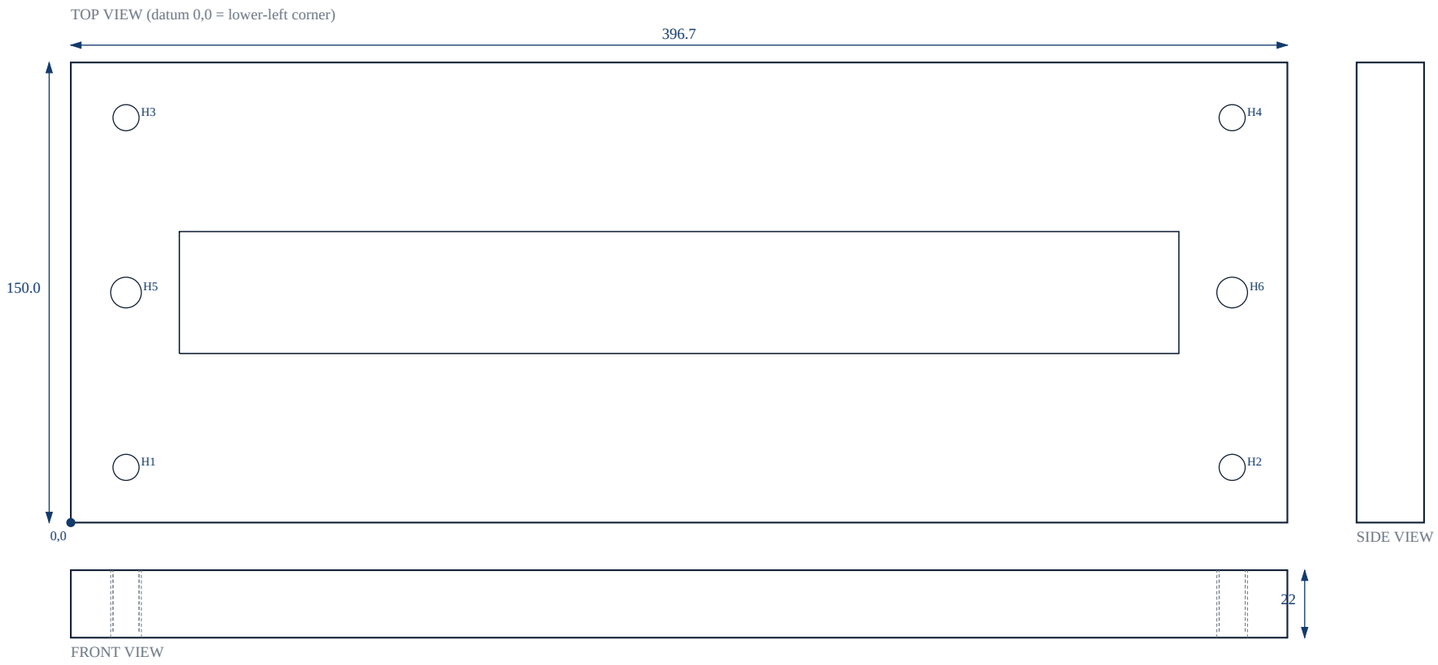
General tolerance	ISO 2768-mK
Projection	
Scale	1:2
Sheet	7 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

07 · Punch holder plate

#	X	Y	Ø	Kind / fit
H1	18.00	18.00	8.5	M10 tapped, depth 20
H2	378.70	18.00	8.5	M10 tapped, depth 20
H3	18.00	132.00	8.5	M10 tapped, depth 20
H4	378.70	132.00	8.5	M10 tapped, depth 20
H5	18.00	75.00	10	Ø10 m6 / H7 dowel, position ±0.01
H6	378.70	75.00	10	Ø10 m6 / H7 dowel, position ±0.01

Punch seat · tolerance m6 / H7 · profile in the DXF (wire cut / CNC)



Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Punch seats m6 / H7

Design to Cost
Drawing no. **EX26-0001-T01-07**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	annealed
Description	Punch holder plate	**Size**	396.70 × 150.00 × 22.00 mm
Item / qty	07 · 1 off	**Make**	-
Material	C45 / S50C	**Finish**	Ra 1.6

General tolerance	ISO 2768-mK
Projection	
Scale	1:2
Sheet	8 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

08 · Backing plate (punch)

#	X	Y	Ø	Kind / fit
H1	18.00	18.00	11	Ø11 screw clearance
H2	378.70	18.00	11	Ø11 screw clearance
H3	18.00	132.00	11	Ø11 screw clearance
H4	378.70	132.00	11	Ø11 screw clearance
H5	18.00	75.00	10	Ø10 m6 / H7 dowel, position ±0.01
H6	378.70	75.00	10	Ø10 m6 / H7 dowel, position ±0.01



Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Thickness ±0.02 mm · 3. Flatness 0.01, parallelism 0.01 · 4. Ground faces Ra 1.6 · 5. Harden and temper to the hardness of the grade; stress relieve before wire cut / EDM finish

Design to Cost
Drawing no. **EX26-0001-T01-08**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	28-32 HRC pre-hardened
Description	Backing plate (punch)	**Size**	396.70 × 150.00 × 10.00 mm
Item / qty	08 · 1 off	**Make**	-
Material	Mould base steel (4140 / 1.7225)	**Finish**	Ra 1.6

General tolerance	ISO 2768-mK
Projection	
Scale	1:2
Sheet	9 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-

Design to Cost · DETAIL DRAWING

09 · Blanking punch

Punch profile = blank – **clearance** · tolerance 0 / –0.01 · profile in the DXF (wire cut / CNC)



Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master. · 2. Profile = blank outline – 0.120 mm per side, 0 / –0.01 mm · 3. Harden and temper to the hardness of the grade; stress relieve before wire cut / EDM finish · 4. Working faces Ra 0.4

Design to Cost
Drawing no. **EX26-0001-T01-09**
Rev **B**

Part (tool for)	EX26-0001 · Die 1: Blanking	**Hardness**	58-62 HRC
Description	Blanking punch	**Size**	325.94 × 39.76 × 60.00 mm
Item / qty	09 · 1 off	**Make**	-
Material	D2 (1.2379)	**Finish**	Ra 1.6

General tolerance	ISO 2768-mK
Projection	
Scale	1:1.5
Sheet	10 / 10
Drawn	Amit Shukla · 2026-10-26
Checked	-
Approved	-